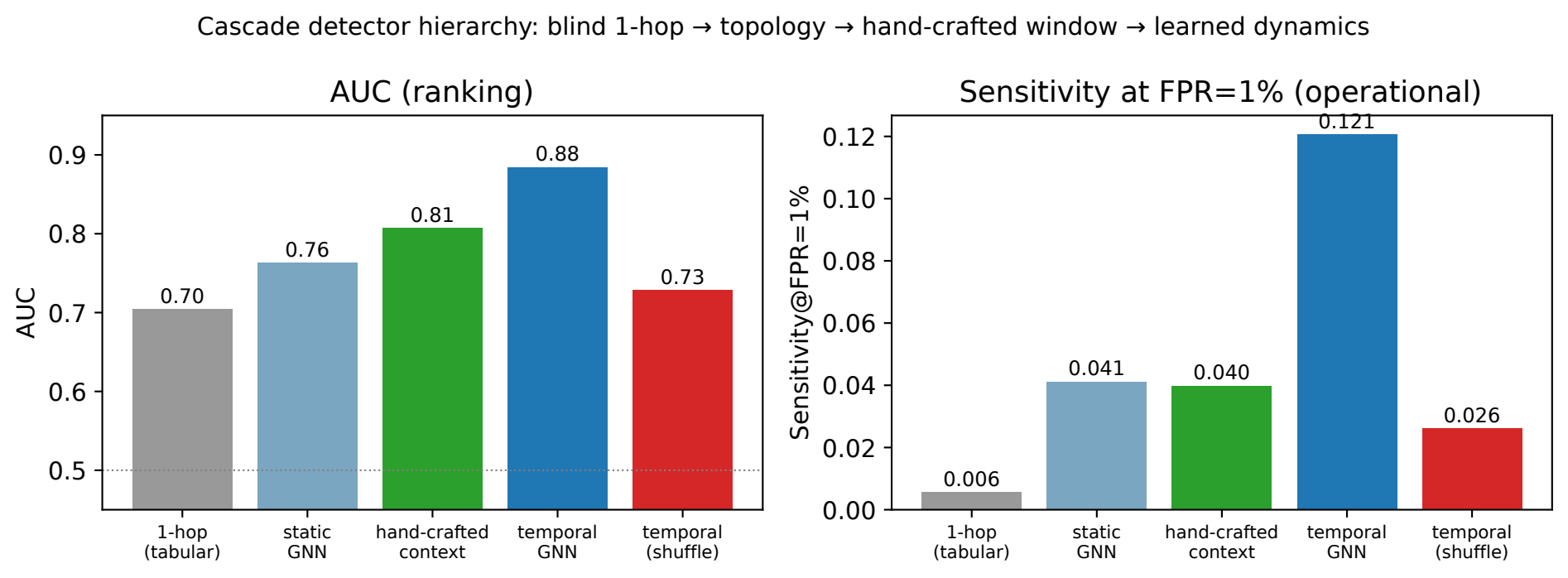
Cascade detector hierarchy: blind 1-hop → topology → hand-crafted window → learned dynamics
AUC (ranking)
AUC
0.9
0.8
0.7
0.6
0.5
0.70
0.76
0.81
0.88
0.73
1-hop (tabular)
static GNN
hand-crafted context
temporal GNN
temporal (shuffle)
Sensitivity at FPR=1% (operational)
Sensitivity@FPR=1%
0.12
0.10
0.08
0.06
0.04
0.02
0.00
0.006
0.041
0.040
0.121
0.026
1-hop (tabular)
static GNN
hand-crafted context
temporal GNN
temporal (shuffle)